Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 2 – Exponentials & Logarithms

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Solve $3^{2x} - 4 \cdot 3^x + 3 = 0$. **[5]**

2. Solve $2^{2x} - 5 \cdot 2^x + 4 = 0$. **[5]**

3. Solve $e^{2x} - 3e^x + 2 = 0$. **[5]**

4. Solve $\log_2(x+1) + \log_2(x-1) = 3$. **[5]**

5. Solve $\log_3(x+2) + \log_3 x = 1$. **[5]**

6. Solve $2\ln x - \ln(x+1) = 0$. **[4]**

7. Solve $\ln(x+1) - \ln(x-1) = 1$. **[5]**

8. Solve $\log_5(2x-1) = 2$. **[3]**

9. Solve $e^x + e^{-x} = 2$. **[4]**

10. Differentiate $y = \ln\left(\frac{2x+1}{x^2+3}\right)$. **[4]**

11. Differentiate $y = \ln(x^2 + 1)$. **[3]**

12. Differentiate $y = e^{3x} \ln x$. **[4]**

13. Solve $3^x = 5$. Give answer in terms of ln. **[3]**

14. Solve $2^{x+1} = 7$. **[3]**

15. Solve $\log_x 16 = 2$. **[3]**

Section B – Hard (Q16–25)

16. Population $P = 500e^{0.05t}$. Initial population, after 10 hours, time when $P = 2000$. **[6]**

17. Population $P = 200e^{0.04t}$. Find when population doubles. **[4]**

18. Solve $e^{2x} - 5e^x + 6 = 0$. **[5]**

19. Solve $\ln(x^2 - 1) = 0$. **[4]**

20. Solve $2\log_2 x + \log_2(x - 1) = 2$. **[6]**

21. Solve $\log_3(x - 1) + \log_3(x + 1) = 1$. **[5]**

22. The curve $y = e^{2x} + e^{-2x}$. Find $\frac{dy}{dx}$ and the coordinates of the stationary point. **[6]**

23. Solve $3^{2x} - 10 \cdot 3^x + 9 = 0$. **[5]**

24. Express $\log_2 12 + \log_2 6 - \log_2 9$ as a single logarithm and evaluate. **[4]**

25. Solve $x^{\log x} = 100x$ (take logs carefully). **[6]**

End of Worksheet • *Show all working clearly.*